

Modeling Two-Dimensional Water Flow and Bromide Transport in a Heterogeneous Lignitic Mine Soil

Uwe Buczko* and Horst H. Gerke

ABSTRACT

Water and solute fluxes in lignitic mine soils and in many other soils are often highly heterogeneous. Here, heterogeneity reflects dumping-induced inclined structures and embedded heterogeneous distributions of sediment mixtures and of lignitic fragments. Such two-scale heterogeneity effects may be analyzed through the application of two-dimensional models for calculating water and solute fluxes. The objective of this study was to gain more insight to what extent spatial heterogeneity of soil hydraulic parameters contributes to preferential flow at a lignitic mine soil. The simulations pertained to the “Bärenbrücker Höhe” site in Germany where previously water fluxes and applied tracers had been monitored with a cell lysimeter, and from where a soil block had been excavated for detailed two-dimensional characterization of the hydraulic parameters using pedotransfer functions. Based on those previous studies, scenarios with different distributions of hydraulic parameters were simulated. The results show that spatial variability of hydraulic parameters alone can hardly explain the observed flow patterns. The measured bromide distributions both in the leachate and the residual concentrations in the soil could not be described at all. Consequently, the observed preferential flow at the site was probably caused by additional factors such as hydrophobicity, the presence of root channels, anisotropy in the hydraulic conductivity, and heterogeneous root distributions. To study the relative importance of these other factors by applying two-dimensional flow models to such sites, the experimental database must be improved, especially with regard to the effects of hydrophobicity and the impacts of high-root-density zones. Single-continuum model approaches may be insufficient for such sites.

MINE SOILS are highly heterogeneous man-made soil systems originating from mine waste materials. Such soils develop on overburden spoil heaps after soil amelioration and forest reclamation (Hüttl and Weber, 2001). Soil heterogeneity is caused by several factors. Overburden sediments of brown coal or lignite mines may contain remnants of lignitic material and different sediments in a spatially distributed manner. As a result of mining sediment masses and dumping technology, spoil heaps consist of an inclined layered sediment structure having widths in the meter-scale with embedded centimeter-sized lignitic fragments (Hüttl and Weber, 2001). Similar two-scale heterogeneity structures containing larger structures with embedded smaller-scale heterogeneities can be found, for instance, in geologi-

cal formations (Heinz et al., 2003) and agricultural soils (i.e., plowing structures containing soil aggregates; Coutadeur et al., 2002).

Knowledge about water and solute flow processes within the thick (up to 100 m) unsaturated zone of the lignitic spoil heaps in the Lusatian Mining District in Germany (Hüttl and Weber, 2001) is essential for estimating water and element balances in those spoil heaps (Gast et al., 1999; Schaaf et al., 1999; Scherzer, 2001) and for monitoring of groundwater quality, which may be impaired by acid mine drainage (Evangelou, 1995; Gerke et al., 1998). Moreover, acidification of the spoil sediments can affect mine soil reclamation and revegetation practices (Hüttl and Weber, 2001).

A common approach for quantification of soil water and element balances is the use of tensiometer and suction cup or other data in conjunction with one-dimensional (1D) vertical numerical modeling (Gast et al., 1999; Schaaf et al., 1999; Scherzer, 2001), using such codes as “SOIL” (Jansson, 1998), for example. Such a 1D approach, however, does not account for the observed considerable spatial variability in soil water tensions (Scherzer, 2001), solute concentrations (Gast et al., 1999; Schaaf et al., 1999), drainage water fluxes (Hangen et al., 2005), the development of finger-type flow paths during infiltration (Hangen et al., 2004), as well as the heterogeneity of soil physical properties of the mineral soil and the spatial distribution of lignitic fragments (Einecke, 2005). The potential effects of spatial heterogeneity on long-term predictions of water and element fluxes in Lusatian mine spoil heaps for two-dimensional (2D) vertical cross sections at a larger (i.e., 10–50 m) scale were previously studied by means of numerical modeling (Gerke et al., 1998; Buczko et al., 2001), but without direct calibration with in-situ measured water and solute fluxes and using only generic spatial distributions of hydraulic parameters.

At this site, three-dimensional (3D) spatially distributed physical and hydraulic parameters of a block of approximately 3 m³ volume were available together with distributed drainage fluxes and bromide-leaching rates and residual tracer distributions (Einecke, 2005; Wecker, 2005). Recently, a 2D-vertical cross section of hydraulic parameters and their small-scale heterogeneity was derived based on soil data of the excavated block (Buczko and Gerke, 2005). The data from this site offer the opportunity for 2D numerical simulations to analyze to what extent spatial heterogeneity of soil hydraulic properties contributes to the observed drainage and leaching patterns.

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Published in Vadose Zone Journal 5:14–26 (2006).

Original Research

doi:10.2136/vzj2005.0003

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Abbreviations: HI, heterogeneity index; SVAT, soil–vegetation–atmosphere–transfer; 1D, one-dimensional; 2D, two-dimensional; 3D, three-dimensional.

Two- or three-dimensional numerical simulations of water flow and solute transport were previously performed, among others, for heterogeneous soils (Roth, 1995; Birkhölzer and Tsang, 1997), hydrophobic soils (Nieber 1996; Ritsema et al., 1998), macroporous soils (Köhne and Gerke, 2005), landfills (Köhler et al., 2001), and spoil heaps (Gerke et al., 1998; Buczko et al., 2001). On the other hand, spatially variable water and solute flow in soils and sediments were experimentally documented by dye tracer experiments (Flury et al., 1994), conservative tracers which were retrieved using spatially distributed suction cups (Roth et al., 1991), cell-lysimeters (Heuvelman and McInnes, 1997; Hangen, 2003; Mylavarapu and Quisenberry, 2005), and three-dimensional soil sampling (Ritsema and Dekker, 1998).

Many of the 2D simulation studies encompassing detailed spatial distributions or elaborated modeling concepts did not provide rigorous comparisons with measured data. For those studies which did give such comparisons, different variables were used in the comparisons between simulations and experiments: Water contents (Nguyen et al., 1999a, 1999b; Pang et al., 2000; Joris and Feyen, 2003); soil water tensions or hydraulic heads (Pang et al., 2000; Joris and Feyen, 2003); depth of groundwater table (Abbaspour et al., 2001; De Vos et al., 2002; Joris and Feyen, 2003); solute concentrations in leachates retrieved by suction cups (Pang et al., 2000); resident solute concentrations after excavating a soil block (Nguyen et al., 1999a); discharge rates in tile drains and corresponding solute concentrations (Abbaspour et al., 2001; Köhler et al., 2001; De Vos et al., 2002; Köhne and Gerke, 2005); finger flow velocities and finger widths in repacked hydrophobic soils (Nieber et al., 2000); spatially distributed drainage rates (Schmalz et al., 2003) and breakthrough curves (Ju and Kung, 1997) below repacked soil containers. In none of these studies, however, simulation results were compared with spatially distributed drainage and leaching rates. On the other hand, cell-lysimeter data reported in the literature (e.g., Strock et al., 2001; Mylavarapu and Quisenberry, 2005) have, to our knowledge, not been analyzed by numerical simulations.

The objective of this study was to gain insights to what extent two-scale spatial heterogeneity of soil hydraulic parameters at a lignitic mine soil site contributes to the observed flow and transport patterns. Using two-dimensional vertical simulation scenarios with different spatial distributions of hydraulic parameters, the resulting simulated water and solute fluxes are compared with in situ fluxes determined experimentally with a cell lysimeter. Differences between simulations and measurements will be used to discuss possible effects of other soil properties and model descriptions.

MATERIALS AND METHODS

Study Site and Experiments

The spoil heap “Bärenbrücker Höhe” (51°49' N, 14°28' E, 99 m above sea level), 12 km northeast of the city of Cottbus in the Lusatian Lignite Mining District was created in 1977 as a preparatory spoil heap for the neighboring lignite open pit

mines “Cottbus Nord” and “Jänschwalde.” The clastic overburden sediments of tertiary and quaternary age were transported with conveyor belts and dumped with spreaders. This process was in contrast to the mining and dumping technology employed in the Lusatian district for regular spoil heaps, which involved overburden conveying bridges (Leibiger, 1964; Matschak 1969). Consequently, for the spoil heap Bärenbrücker Höhe, the inclined spoil sediment layers have a thickness of only 40 cm on average. The relatively low dumping heights resulted in lower bulk densities within the impact zones of the dumped spoil sediment flux, in comparison with regular spoil heaps (Matschak, 1969).

The site was ameliorated in 1978 with 190 t ha⁻¹ calcium oxide (CaO) in the form of power plant flue ash, which was incorporated into the upper 40 cm soil horizon by plowing. Afforestation ensued in 1982 with 7375 trees ha⁻¹ of *Pinus nigra* the roots of which are mainly restricted to the upper ameliorated 40-cm soil depth (Hangen et al., 2004). Fine sand (0.063–0.2 mm particle diameter) is the predominant texture class (Neumann, 1999). The soil type is an Anthropic Regosol according to the World Soil Classification (FAO/ISSS/ISRIC, 1998). The content of residual lignite fragments, with sizes ranging from dust to more than 5 cm in diameter, averages 17 vol.-% (Einecke, 2005).

For the period 1996 to 1999, mean annual precipitation amounted to 850 mm (Scherzer, 2001). This value exceeded the long-term (1951–1980) mean value of about 580 mm yr⁻¹ recorded at the meteorological station of Cottbus (Veit et al., 1987).

Water and element budgets for this site were evaluated at a different plot nearby, for the period 1995 to 1999, based on tensiometer and time domain reflectometry (TDR) measurements and 1D simulations (Schaaf et al., 1999; Gast et al., 1999; Scherzer, 2001). In 2000 and 2001, drainage water fluxes at 110-cm soil depth were measured in situ by means of a cell-lysimeter device (Hangen et al., 2005). Using these data, it could be demonstrated, that the assumption of 1D-vertical flow and transport was invalid, by comparing one-dimensional model predictions of drainage rates (about 4–5 mm yr⁻¹) obtained with previously calibrated mine soil hydraulic parameters (Scherzer, 2001) with those directly measured with the cell lysimeter (31.6 mm for the Year 2001). Further, in the course of the experiment, a multiple tracer mix was applied (Hangen et al., 2005): on 7 Nov. 2000, 0.005 kg m⁻² Bromide was applied using a 1.6 × 10⁻³ m³ m⁻² (= 1.6 l m⁻²) tracer solution followed by 4 × 10⁻³ m³ m⁻² (= 4 l m⁻²) distilled water. The procedure took approximately 6 h.

After the end of the cell-lysimeter experiment (i.e., 21 Sept. 2001), the soil block above the lysimeter cells was excavated and soil physical parameters (texture, bulk density, lignite contents) were measured for single soil blocks of 27-cm edge length (Hangen, 2003; Einecke, 2005; Wecker, 2005). Moreover, images of several vertical cross sections through the soil block were used for determining the spatial distribution of inclined spoil layers and lignite fragments. Based on these spatial patterns and the measured soil physical parameters, hydraulic parameters of the different structural regions of the spoil cross section were estimated (Buczko and Gerke, 2005).

Modeling

For simulating water flow and nonreactive solute transport, the HYDRUS-2D program (Simunek et al., 1999) was used. HYDRUS-2D describes variably saturated water flow with the Richards equation and conservative (nonsorbing, nonreactive) solute transport with the convection–dispersion equation.

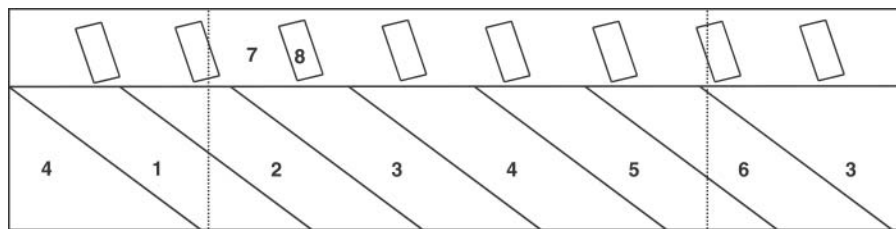


Fig. 1. Sketch of simulated 2D vertical cross section with numbers of the structural units. Width: 450 cm. Height: 110 cm. Dotted vertical lines denote the boundaries of the excavated cross section.

The soil hydraulic functions were described according to van Genuchten (1980). The water retention function, $\theta(\psi)$ was given by

$$\theta(\psi) = \begin{cases} \theta_r + \frac{\theta_s - \theta_r}{(1 + |\alpha_{vG}\psi|^n)^m} & \psi < 0 \\ \theta_s & \psi \geq 0 \end{cases} \quad [1]$$

where θ_s is the saturated ($L^3 L^{-3}$) and θ_r the residual water content ($L^3 L^{-3}$), α_{vG} (L^{-1}) and n are empirical parameters and $m = 1 - 1/n$. The hydraulic conductivity function, $K(\psi)$, was described (Mualem, 1976) as:

$$K(\psi) = K_s \frac{\left[1 - (\alpha_{vG}\psi)^{n-1} [1 + (\alpha_{vG}\psi)^n]^{-m}\right]^2}{[1 + (\alpha_{vG}\psi)^n]^{m/2}} \quad [2]$$

The basis for the simulated scenarios was a cross section through the excavated soil monolith of 250-cm width and 110-cm height. The estimation procedure and values of the hydraulic parameters are given in Buczko and Gerke (2005). In Fig. 1, the cross section for the simulation is depicted schematically. The simulated flow domain was 450 cm in the horizontal and 110 cm in the vertical. The 100-cm extra width on each side as compared to the excavated spoil profile was introduced to eliminate boundary effects in the simulations. The finite element grid used in the simulations was irregular, with a mean discretization of approximately 4 cm. The total number of nodes was 4950 and the number of triangular sub-elements 9519.

The soil water initial conditions were obtained by means of a one-dimensional simulation with the soil-vegetation-atmosphere-transfer (SVAT) model COUP (Jansson and Karlberg, 2001) over the period 26 May 1995 to 31 Oct. 2000 (i.e., the time before the start of the 2D simulations). Rates of precipitation, evaporation, and transpiration with a temporal resolution of 1 h were used as upper boundary condition for water flow. The transpiration and evaporation rates were obtained from simulations with the COUP-model using the Penman-Monteith-equation. The lower boundary conditions for water flow were chosen such as to reproduce the actual conditions in the field experiment (Hangen et al., 2005): For the most of the time, no suction was applied at the suction plates. The corresponding boundary condition in the simulations was a no-flow condition. At intervals of 2 to 5 d (depending on the rainfall conditions) a suction of approximately 30 cm of water was applied at the plates for 30 to 60 min. Accordingly in the modeling, for the dates when water was extracted in the field, a prescribed pressure head of -30 cm water lasting 1 h was used. The 2D simulations started 1 Nov. 2000 and ended 21 Sept. 2001 (i.e., the simulated period was 324 d).

For the simulated two-dimensional scenarios, we tried to reproduce the appreciable observed heterogeneity in measured water and solute fluxes primarily by accounting for spatial variability in the hydraulic parameters. Four differ-

ent scenarios of spatially variable soil hydraulic properties were considered:

- Homogeneous distribution of soil hydraulic parameters. The hydraulic parameters were calculated as area-averaged means of the hydraulic parameters of Scenario C, which considered several homogeneous spoil layers.
- Geostatistical scaling of the soil hydraulic parameters: The natural logarithm of the scaling factor for the hydraulic conductivity, α_{rK} , was assumed to vary in space with a variance of 1.0 and a correlation length of 30 cm.
- Different, internally homogeneous spoil layer structures; the hydraulic parameters for the different structural units are compiled in Table 1. They were estimated with pedo-transfer functions based on measured bulk density, texture, and lignite contents (Buczko and Gerke, 2005).
- Spoil structures with small-scale variability in hydraulic parameters; the same spoil structures as in case C were combined with small-scale variability in the hydraulic parameters (variance 1.1; correlation length 10 cm). Methods for estimating the small-scale variability are described in Buczko and Gerke (2005). In essence, the estimation procedure was based on different area fractions of lignite fragments within 100 to 200 cm² subzones of the excavated 27- by 27-cm mine soil cross sections.

The geostatistical distributions of the scaling factors in cases B and D were generated using a sequential Gaussian algorithm, as implemented in HYDRUS-2D. We assumed self-similarity in pore space according to Miller and Miller (1956) scaling. For case B, three realizations were evaluated, while for case D, we used five realizations. Case C provided a reference for studying other scenarios involving anisotropy of the hydraulic conductivity, wedge-shaped roots at boundaries between inclined spoil ribs, root distribution according to measured block data, temporally averaged infiltration rate, and seepage faces between the suction plates at the lower boundary.

Table 1. Hydraulic parameters of the different structural units of the simulated cross section. Units 1 through 8 refer to Scenarios B, C, and D (Fig. 1). The last two lines refer to Scenario A.

Unit	K_s	θ_s	θ_r	n	α_{vG}
	cm d ⁻¹				cm ⁻¹
1	103.0	0.587	0.048	1.403	0.034
2	73.0	0.517	0.067	1.426	0.052
3	266.4	0.540	0.037	1.378	0.053
4	9.0	0.517	0.102	1.550	0.025
5	17.1	0.493	0.104	1.752	0.017
6	2.6	0.446	0.123	1.449	0.030
7	250	0.582	0.177	1.935	0.030
8	250	0.468	0.209	1.888	0.021
Scenario A, surface layer	250	0.562	0.182	1.92	0.029
Scenario A, soil depth below 40 cm	26.2	0.514	0.082	1.473	0.032

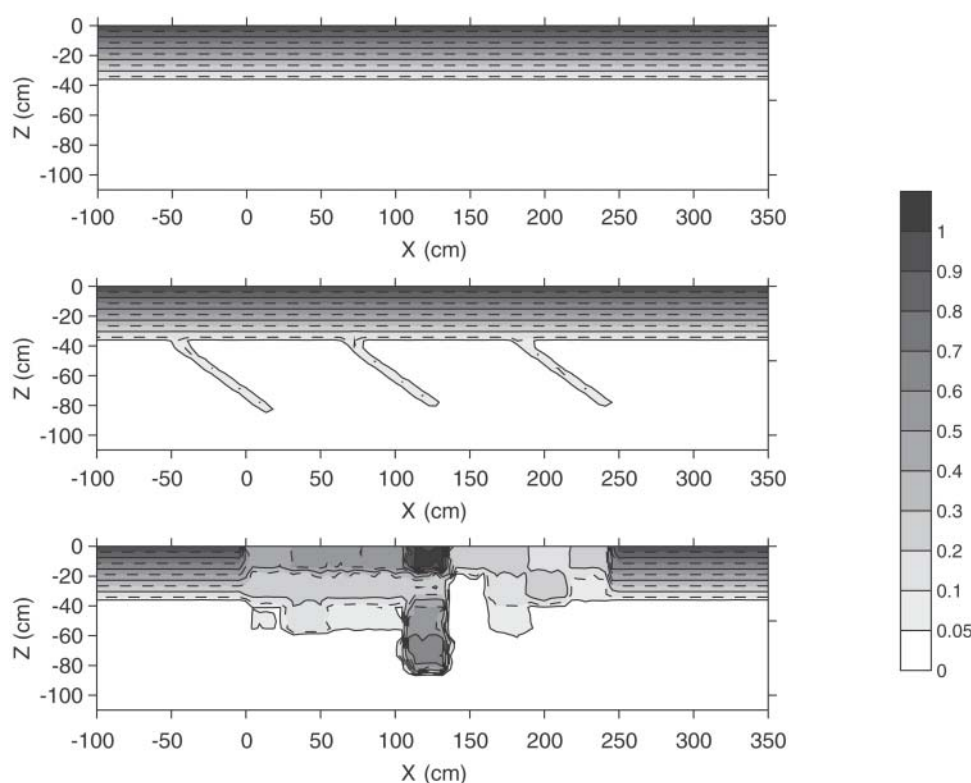


Fig. 2. Root density distributions (dimensionless) used in the simulated scenarios. For the standard scenario (top), a horizontally homogeneous distribution is linearly decreasing from the soil surface toward 40-cm depth. The scenario with wedge-shaped insertions of roots between the inclined spoil layer structures (middle) has additionally narrow seams of roots parallel to the spoil layers down to 90-cm soil depth. The root distribution scenario at the bottom uses measured root density data.

For the standard simulation scenario, the following conditions were assumed: isotropy in the hydraulic properties, no hysteresis in hydraulic functions, a homogeneous root distribution in the horizontal direction, but linearly decreasing from the soil surface to a 40-cm depth. The different root distributions are depicted in Fig. 2, whereas 2D distributions of the hydraulic parameters for the Scenarios B, C, and D are shown in Fig. 3. For the simulated scenario with anisotropy in the hydraulic conductivity, parallel to the dip of the layered spoil structures, the hydraulic conductivity was multiplied by a factor of 2.0, whereas perpendicular to those layers this factor was 0.5.

RESULTS AND DISCUSSION

Simulated Drainage Rates

Spatially integrated infiltration and measured non-capillary and suction-cell drainage rates are depicted as daily values for the entire simulated period in Fig. 4. Simulated water fluxes at the bottom of the spoil cross section for the four different spatial distributions of soil hydraulic properties are depicted as daily values in Fig. 5.

The total cumulated simulated bottom water flux was highest for Scenario D₁ (43 mm), while for the other scenarios the cumulative flux was approximately equal to the cumulated measured flux (31.6 mm, Table 2). Measured daily values of drainage were highest in the first half of March, whereas the simulated daily bottom water flux for all scenarios was highest in the second half of March and the first half of April. The highest measured daily flux rates were almost 3 mm d⁻¹, whereas the

simulated daily bottom water fluxes in no case exceeded 1.5 mm d⁻¹. These discrepancies between measured and simulated drainage rates indicate, that the degree of preferential flow measured in the field experiments could not be accounted for by any of the simulated scenarios. Although in September 2001, approximately 10 mm of seepage water was collected in the field, no bottom water flux was simulated in any of the scenarios for this month.

Three different realizations of the purely geostatistical spatial variability scenario (B) revealed rather similar temporal patterns of simulated daily values of the water flux (not shown here), whereas the cumulative fluxes differed significantly (Table 2). Similarly, the five different realizations of the scenario with spoil structures and small-scale variability of hydraulic parameters (D) exhibited no marked differences in daily-simulated bottom water fluxes, but somewhat clearer differences in the cumulative fluxes (Table 2).

The difference in the effects of anisotropy in the hydraulic conductivity and of a spatially heterogeneous root distribution on simulated average bottom water fluxes was small and barely discernible (results not shown here). Insertion of centimeter-wide seepage faces between the suction cells in the simulations yielded no water fluxes at those seepage face boundary parts. Consequently, water fluxes were diverted from the seepage faces toward the suction cells. Spatially averaged, however, hardly any differences occurred in the simulated bottom water fluxes

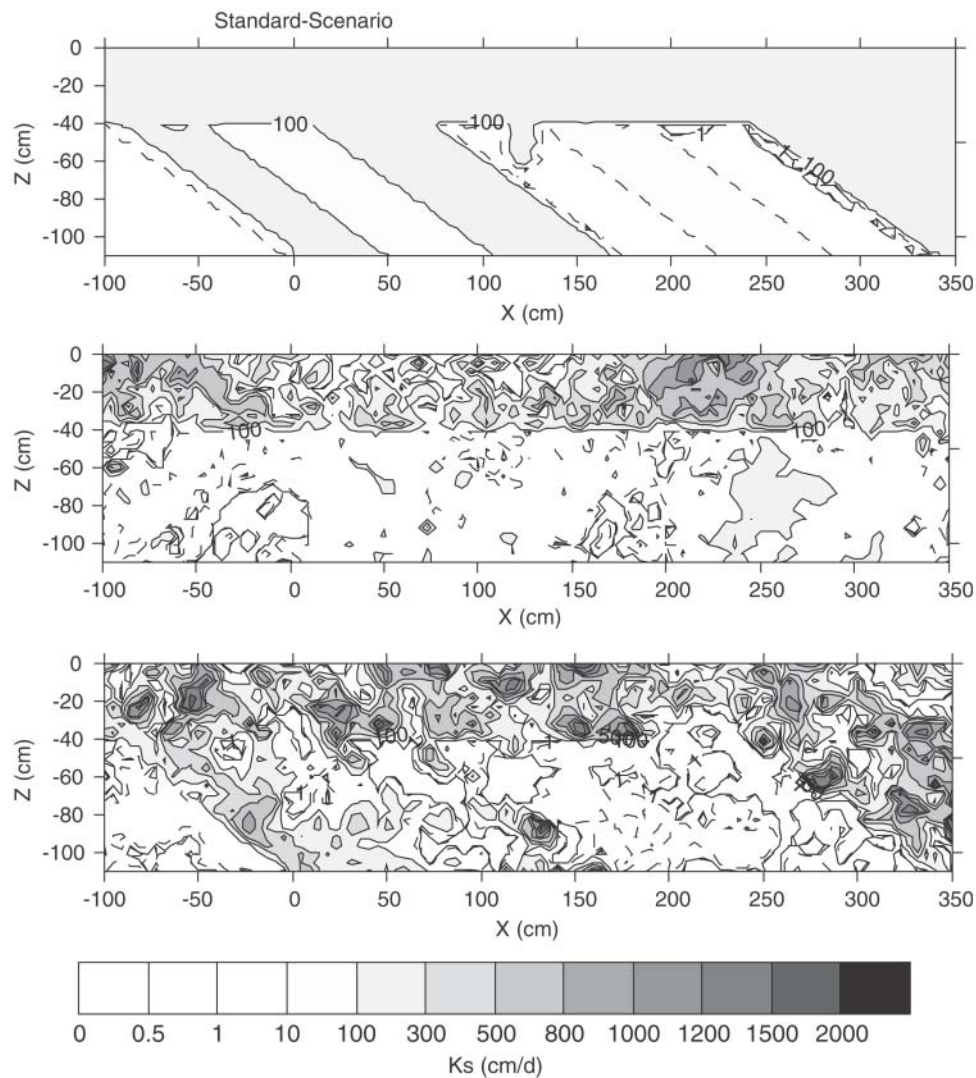


Fig. 3. Two-dimensional distribution of K_s for the standard scenario (C, top), for one realization of the geostatistical simulation scenario (B, middle), and for the small-scale variability scenario (D, bottom).

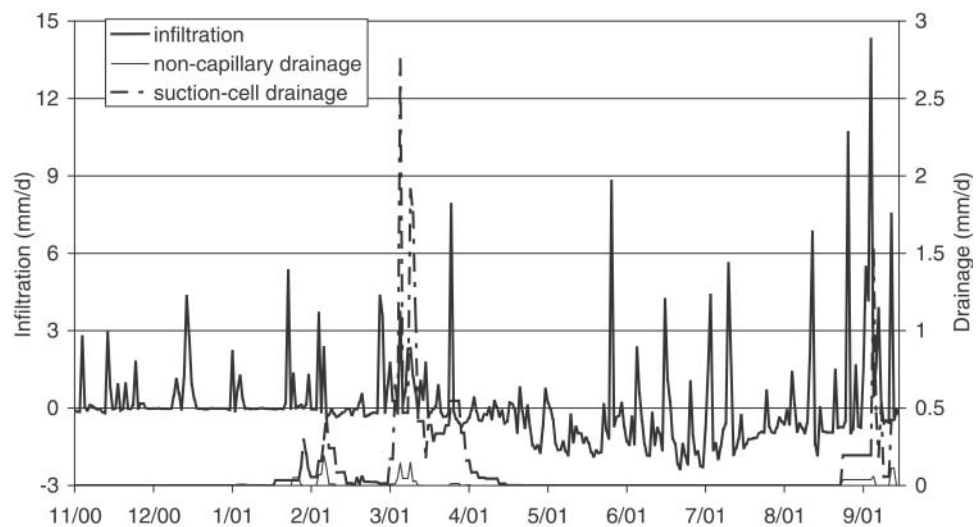


Fig. 4. Spatially-averaged daily values of measured water fluxes at the soil surface (infiltration–left axis) and at the 110-cm soil depth (drainage–right axis) for the period from 1 Nov. 2000 to 21 Sept. 2001.

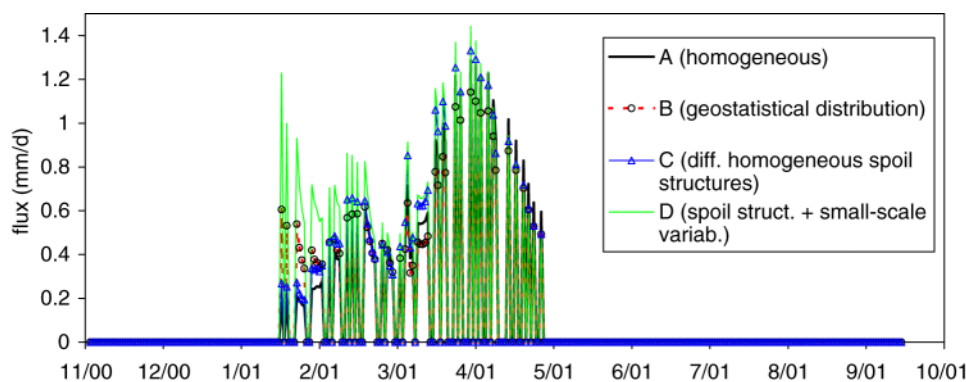


Fig. 5. Spatially-averaged daily values of simulated fluxes at 110-cm soil depth for the period from 1 Nov. 2000 to 21 Sept. 2001. (Scenarios with different distributions of soil hydraulic properties, horizontally homogeneous root distribution.)

as compared to the standard scenario without those seepage faces. In the field measurements, however, the water retrieved by those seepage faces amounted to 2 mm (i.e., 6% of the total cumulative measured flux of 31.6 mm). To retrieve seepage water through a seepage face boundary, the matric potential above that boundary has to be positive (i.e., the soil must be water saturated). In contrast to the field measurements, where the matric potential necessarily must have been positive at least for short periods, this never occurred in the simulations. This suggests the presence of fast-reacting preferential flow processes, which were not captured by the simulations.

Spatial Distribution of Water Fluxes

The distribution of cumulative water fluxes along the bottom boundary is depicted in Fig. 6 for the four different spatial distributions of the soil hydraulic parameters. Clearly, the homogeneous scenario exhibited no spatial heterogeneity in the simulated bottom water fluxes. While the scenario with purely geostatistical variation (B) showed spatial variations in the bottom water fluxes, this variability had little resemblance with the spatial distribution of measured fluxes. The different re-

alizations of this scenario exhibited distinct differences (results not shown here), but none showed similarities with the measured patterns in the bottom water flux.

Both Scenario C having different internally homogeneous spoil structures and Scenario D having additional small-scale variability exhibited spatial patterns in simulated water fluxes somewhat similar to the measured patterns, except that the extreme spatial heterogeneity in measured drainage water fluxes was not reproduced by any of the scenarios.

The calculated spatial variability in simulated bottom water fluxes for five different realizations of the small-scale variability scenario (D) showed pronounced differences among the realizations (Fig. 7). Whereas the general spatial pattern of the measured bottom water flux was reproduced by most realizations, the strong heterogeneity in measured fluxes, especially the high fluxes in the suction cell at the center, could not be reproduced well with any of the realizations.

The effect of anisotropy in hydraulic conductivity on spatial variability of simulated bottom water fluxes was relatively small but still clearly discernible (not shown graphically). This effect may be caused by the enhanced channelling of water flow parallel to the inclined spoil

Table 2. Cumulative measured and simulated water fluxes for different periods and different simulation scenarios.

		Water flux		
		Spring 2001		Late summer 2001
		7 Nov. 2000–21 Sept. 2001	(1 Jan.– 1 May 2001)	(28 Aug.– 21 Sept. 2001)
		mm		
Measured	Throughfall	211	67.3	55.5
	Infiltration	33.5	36.6	43.1
	Suction-cell drainage	29.6	22.6	7.0
	Free gravity drainage	2.0	1.3	0.7
Simulated capillary drainage rates	Homogeneous	A	32.7	—
	Geostatistical spatial variability	B ₁	31.8	—
		B ₂	25.8	—
		B ₃	36.9	—
	Standard	C	34.3	—
	Spoil structures with small scale variability of hydraulic parameters	D ₁	43.0	—
		D ₂	40.3	—
		D ₃	40.5	—
		D ₄	40.8	—
		D ₅	36.5	—
	Anisotropy of hydraulic conductivity	C	34.9	—
	Seepage face between suction plates	C	34.0	—
	Root distribution according to measured block data	C	34.1	—
	Root distribution: measured block data and spoil structures with small- scale variability	D ₅	36.3	—

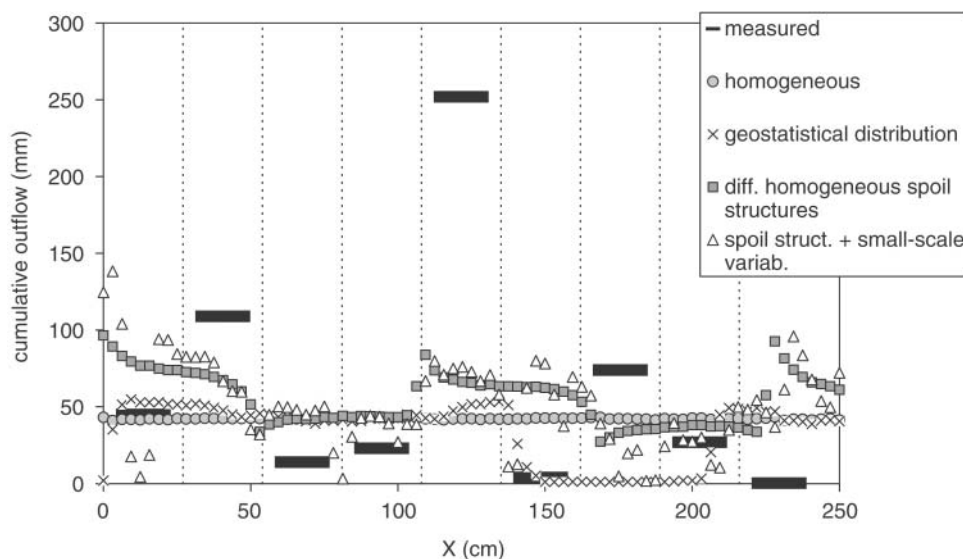


Fig. 6. Spatial variability in simulated and measured cumulative water fluxes at the 110-cm soil depth (Scenarios with different distributions of soil hydraulic properties). Dashed vertical lines denote the boundaries of the suction cells.

layers when assuming anisotropy of hydraulic conductivity. Insertion of narrow seepage faces between the suction plates had a distinct influence on the simulated bottom water fluxes (not shown graphically). No water was simulated to flow through the seepage faces themselves, which means, that the water was diverted toward the neighboring suction plates and enhanced the flow there. The root distribution was found to have little or no influence on the simulated bottom water fluxes, even with small-scale variability in the hydraulic properties.

Spatial heterogeneity in water flow within the soil block may be assessed and quantified by plotting the cumulative fraction of the total water flow vs. the cumulative area fraction contributing to this flow (sorted in descending order) (Quisenberry et al., 1994; Stagnitti et al., 1999). A straight diagonal line through such a plot marks a spatially homogeneous flow pattern, whereas deviations from this 1:1 line indicate spatially heteroge-

neous flow. The more the line deviates from the 1:1 diagonal, the more pronounced the spatial heterogeneity in the water flow pattern. Homogeneity of water flow is seen in Fig. 8 for the scenario with a homogeneous distribution of soil hydraulic parameters (A), whereas the scenarios with heterogeneity in the soil hydraulic parameters exhibited a slight heterogeneity in water flow. The heterogeneity in cumulative measured water fluxes is, on the other hand, much more pronounced. Using the same evaluation method as in Fig. 8, different realizations of geostatistically generated distributions produced distinctly different patterns of water flow heterogeneity. Assuming anisotropy in the hydraulic conductivity, the spatial heterogeneity in water flow is slightly enhanced. Different root distributions and temporally averaged upper boundary conditions did not enhance the spatial heterogeneity in water flow. On the other hand, inserting seepage faces between the

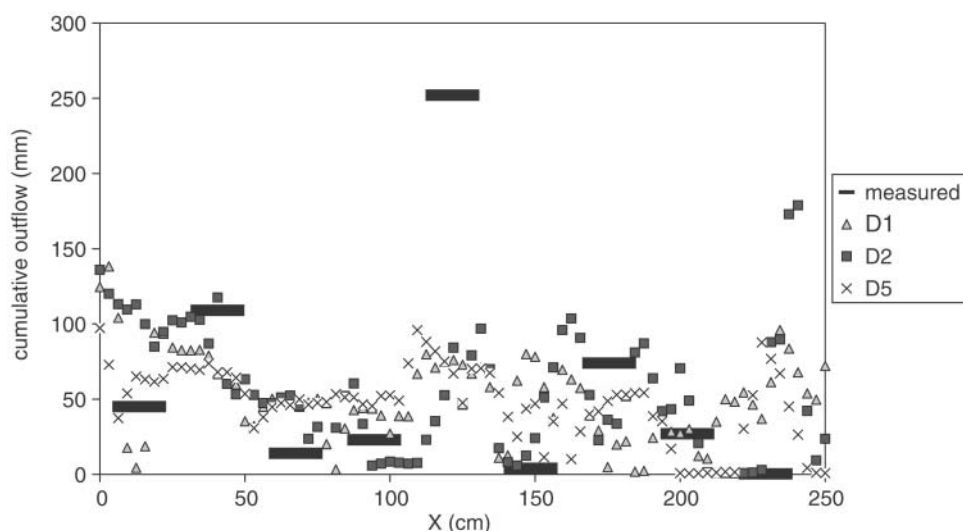


Fig. 7. Spatial variability in simulated and measured cumulative water fluxes at the 110-cm soil depth (Different realizations of the small-scale variability scenario (D)). Dashed vertical lines denote the boundaries of the suction cells.

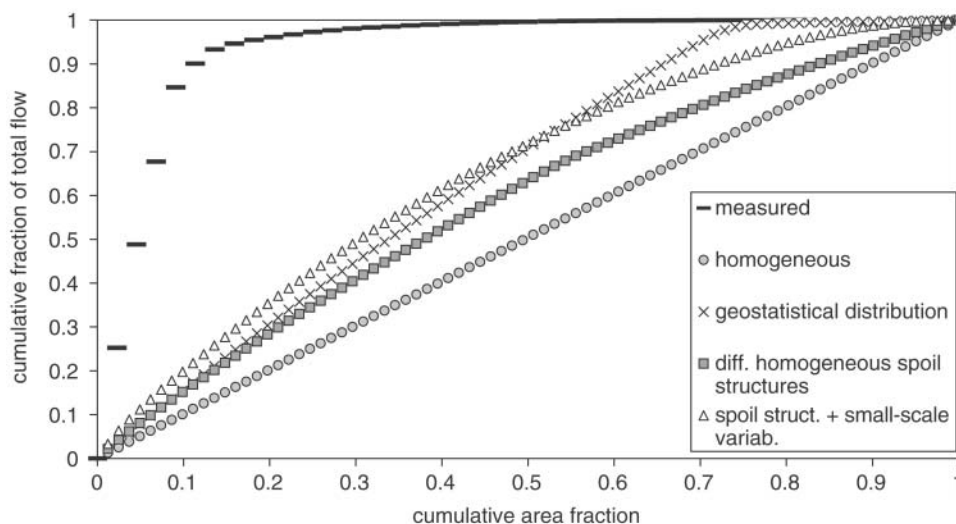


Fig. 8. Cumulative fraction of total water fluxes as a function of the cumulative cross sectional area fraction (Scenarios with different distributions of soil hydraulic properties).

different cells of the lysimeter clearly enhanced the spatial heterogeneity in water flow (not shown here).

According to Stagnitti et al. (1999), spatial heterogeneity in water flow data such as those plotted in Fig. 8 may be quantified by fitting a cumulative β function to the data. A heterogeneity index (HI) can then be defined as

$$HI = \frac{\sqrt{3\sigma_x}}{\mu_x} = \sqrt{\frac{3\zeta}{\alpha(\alpha + \zeta + 1)}} \quad [3]$$

where σ_x denotes the standard deviation and μ_x the mean value of the β distribution function; α and ζ are the two parameters of the β function (see Stagnitti et al. (1999), for details).

In Table 3, the heterogeneity indices for water flow are compiled for the simulated scenarios and the cumulated measured data. The heterogeneity indices of the simulations are distinctly lower than those for the measured data. Stagnitti et al. (1999) reported for different,

Table 3. Values of heterogeneity index (HI) for the different simulated scenarios and the measured water flux (calculated according to Stagnitti et al., 1999).

Scenario	Spatial distribution of hydraulic parameters	HI
Homogeneous	A	1
Geostatistical spatial variability	B ₁	1.195
	B ₂	1.109
	B ₃	1.194
	C	1.164
Standard Spoil structures with small-scale variability of hydraulic parameters	D ₁	1.253
	D ₂	1.314
	D ₃	1.287
	D ₄	1.343
	D ₅	1.186
Anisotropy of hydraulic conductivity	C	1.207
Infiltration rate temporally averaged	C	1.174
Seepage face between suction plates	C	1.221
Root distribution according to measured block data	C	1.168
Root distribution according to measured block data and spoil structures with small-scale variability of hydraulic parameters	D ₅	1.185
Cumulative measured water fluxes		1.634

mostly loamy-textured soils in Australia values for HI ranging between 1.31 and 1.89. They used soil blocks of 30 to 40 cm on each side and applied artificial irrigation at rates between 24 and 192 mm d⁻¹. Percolation water was retrieved with multiple wick lysimeters within individual areas of 6 × 6 cm². In another study, de Rooij and Stagnitti (2000) reported a heterogeneity index of 1.32 (for solute transport) for a dune sand in The Netherlands. The measured cumulative water flow data for the Bärenbrück site yielded HI = 1.634, which is distinctly higher than the values for all the simulated scenarios.

Spatial Distribution of Bromide Concentrations

Bromide was measured in the seepage water for the first time in the beginning of February 2001 (i.e., 90 d

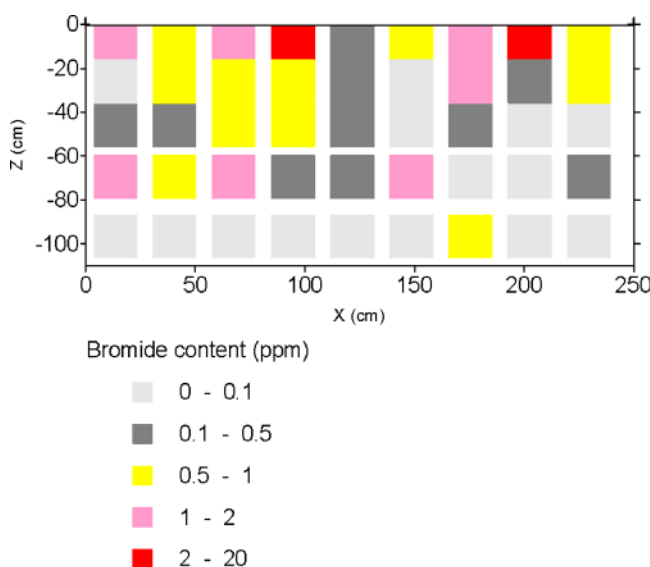


Fig. 9. Measured two-dimensional residual bromide concentration distributions (ppm) after the end of the experiment for the middle vertical soil slice (Y3) (Wecker, 2005). The values refer to the soil bulk dry mass.

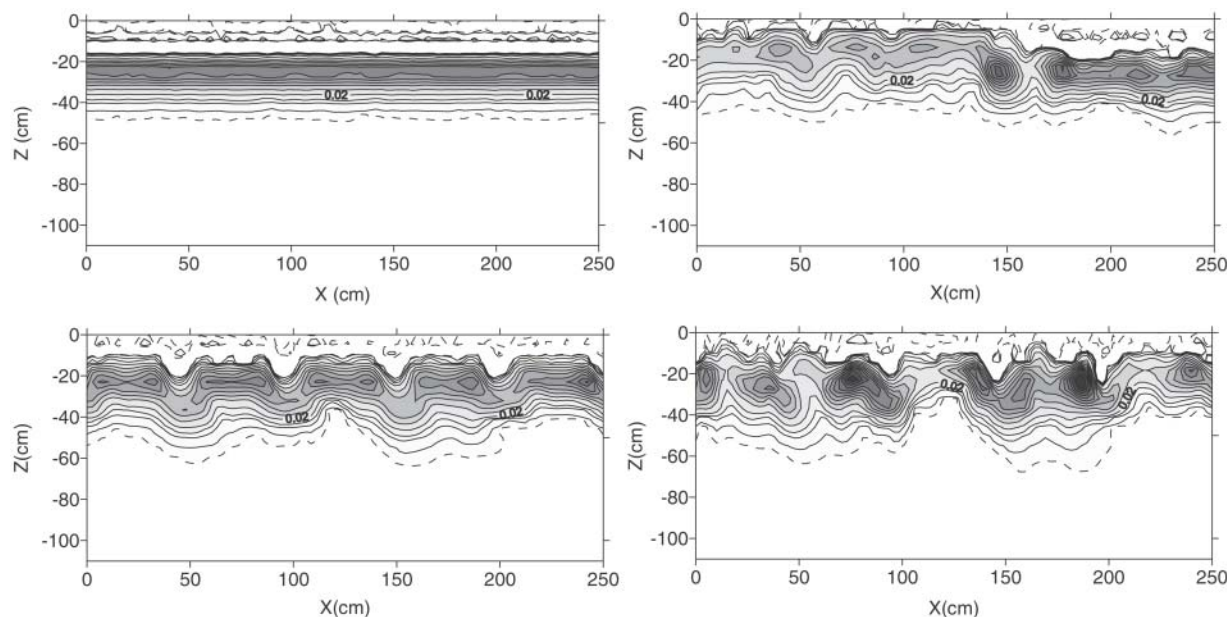


Fig. 10. Simulated two-dimensional bromide concentration (kg m^{-3}) distributions at the end of the simulated period (Scenarios with different spatial distributions of hydraulic parameters). A (= homogeneous, top left), B (top right), C (bottom left), D (bottom right).

after the tracer application). Until the end of the cell-simulator experiment in September 2001, the sum of bromide mass in the seepage water amounted to 384 mg, that is, 2.34% of the applied mass (Hangen, 2003). For all simulated scenarios, on the other hand, no bromide was found at the bottom of the simulated domain during the whole simulated period.

The residual bromide contents which were measured in the excavated soil block displayed a highly heterogeneous pattern with indications of preferential flow parallel to the inclined spoil layer structures (Fig. 9). In contrast, simulated concentrations (shown here for the end of the simulated period at $t = 324$ d, Fig. 10) were clearly highest in the upper third of the flow domain, while hardly any bromide was present below the 40-cm soil depth in the simulations. The simulated two-dimensional distributions are shown in Fig. 10 for the different spatial distributions of hydraulic parameters. For the scenarios with inclined spoil layer structures (C and D), solute transport was clearly channelled parallel to the spoil structures, although the depth distribution of the simulated concentrations was much more shallow in comparison with the measured

bromide concentrations. For all heterogeneous scenarios, a horizontal redistribution and spreading of the solute was apparent. While this is typical for simulated solute transport in heterogeneous domains with transient upper boundary conditions (e.g., Russo et al., 1994), we also observed this in the measured bromide distribution (Fig. 9).

The differences in the concentration pattern among the different realizations of the geostatistical scenarios were relatively large in detail but similar in principle (not shown here). On the other hand, there is scarcely any visible effect of hydraulic conductivity anisotropy on the simulated 2D distributions of bromide concentrations, in contrast to the deviation of the water fluxes from the vertical, compared to the standard scenario without local-scale anisotropy in hydraulic conductivity (see section entitled Spatial Distribution of Water Fluxes). One reason for this discrepancy could be that the anisotropy was assumed to be restricted to the soil region below the 40-cm depth, since ameliorative plowing had disturbed the upper 40 cm. Obviously, only a minor part of the solute mass was transported below 40 cm in the simulations and could possibly be influenced

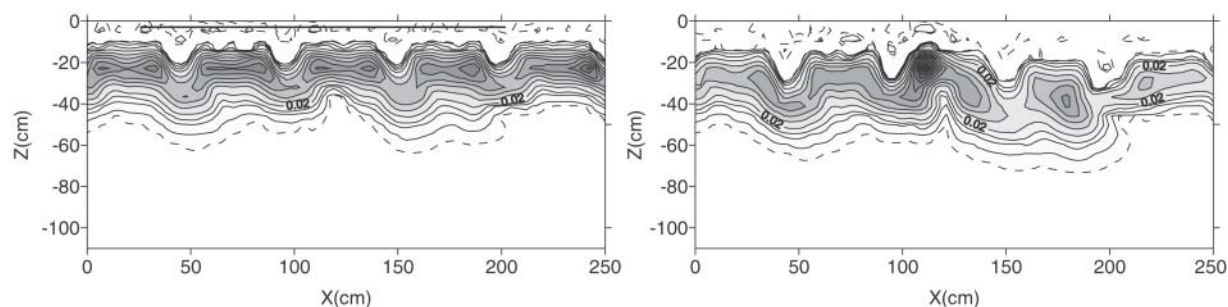


Fig. 11. Simulated two-dimensional bromide concentration (kg m^{-3}) distributions at the end of the simulated period. Effect of root distribution (left: Horizontally homogeneous; right: Measured root distribution).

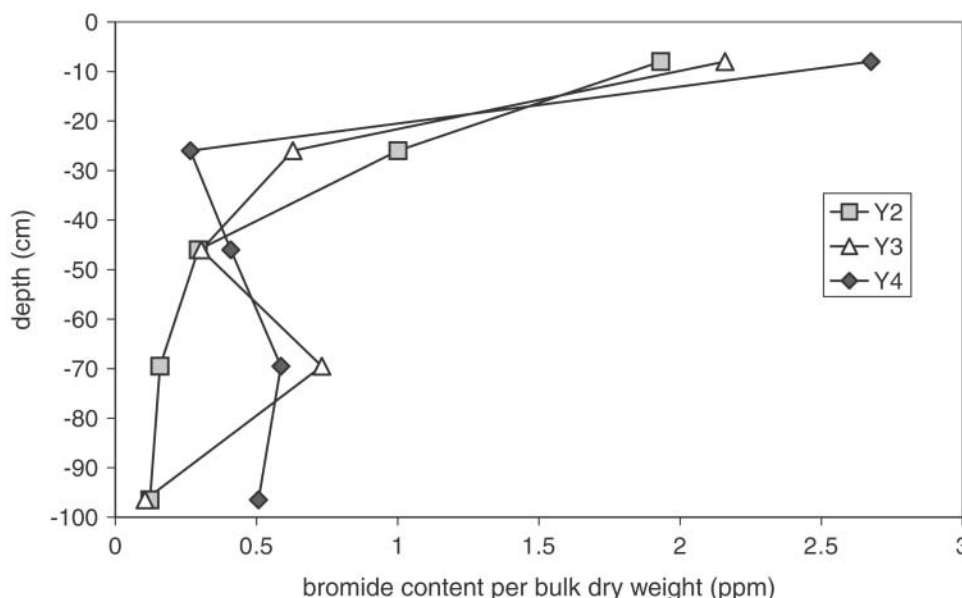


Fig. 12. Horizontally averaged measured bromide contents (ppm) for different vertical soil cross sections (Y2, Y3, and Y4).

by the anisotropy of hydraulic conductivity. Summarizing, the effect of local-scale anisotropy in the hydraulic conductivity on the simulated water and solute fluxes was rather small, although the anisotropy ratio of 4 we had used was higher than values reported for unconsolidated eolian (1.25–1.35; van den Berg and de Vries, 2003), or litoral (1.33–1.57; Burger and Belitz, 1997) sediments.

Using measured block data for the root distribution, instead of a horizontally homogeneous root distribution, had a distinct effect on the simulated two-dimensional concentrations (Fig. 11). Concentrations in the central part of the simulated domain were higher than in the surrounding region, while solute displacement seemed

somewhat retarded. Since this central part clearly exhibited the highest measured root mass, the higher root water extraction and concomitant upward water movement during the vegetation period may explain the concentration “hot spot” there. However, the measurements showed relatively low bromide concentrations in the central part. On the other hand, the measured bromide mass in the seepage water below this central part was distinctly higher than in the other cells (Hangen, 2003). Measured bromide in the center was essentially restricted to February and March 2001 (i.e., before the vegetation period and the onset of transpiration). The roots in the central zone likely acted as channels for preferential downward flow in the winter months and as transpiration

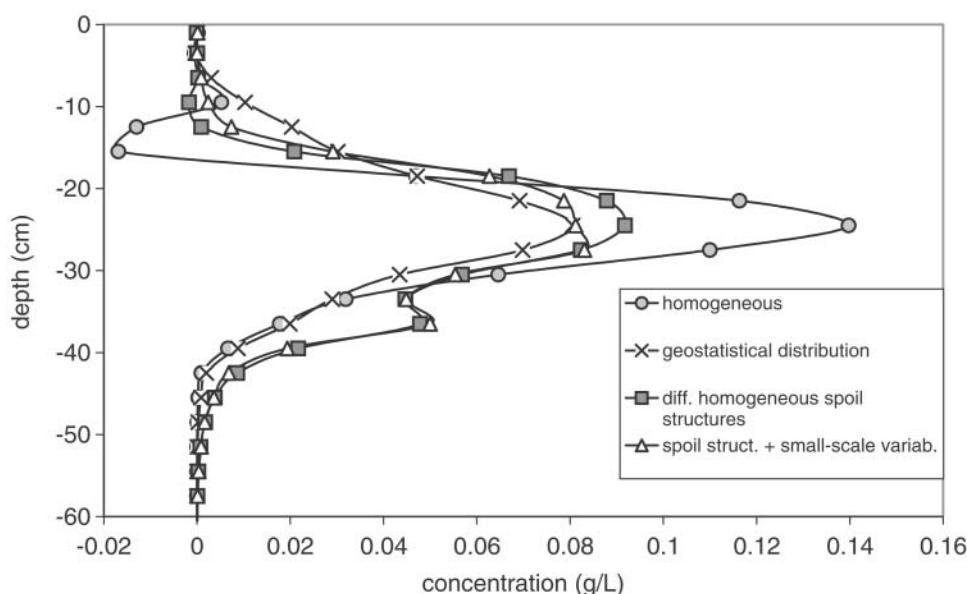


Fig. 13. Horizontally averaged simulated concentrations at the end of the simulated period (Scenarios with different distributions of soil hydraulic parameters).

conduits during the vegetation period. This was not accounted for, however, in the simulations.

Horizontally Averaged Concentrations

Horizontally averaged simulated concentrations allow a somewhat integrated assessment of the solute transport process. In the following, horizontally averaged simulated concentrations are given for $t = 324$ d (i.e., the end of the simulation period).

Measured horizontally averaged bromide contents (Fig. 12) were clearly highest in the upper soil layer (0–20-cm soil depth), and then exhibited a rapid decline vs. depth, and a weaker, secondary maximum at the 70-cm depth. In contrast, simulated horizontally averaged concentrations showed, similarly for all four distributions of soil hydraulic parameters, maximum values at about the 25-cm soil depth (Fig. 13). Moreover, for the two scenarios with inclined spoil layer structures (C and D), a secondary concentration peak between 35 and 40 cm depth was visible.

Anisotropy in hydraulic conductivity had almost no visible effect on horizontally averaged simulated concentrations. The effect of horizontally heterogeneous measured root data was enhanced spreading of the horizontally averaged concentration distribution. Narrow seepage faces between the suction plates had practically no effect on the simulated horizontally averaged concentrations. Measured block data of root distribution in the case of small-scale variability induced, similarly as for homogeneous spoil layer structures, a spreading of the concentration front (not shown here).

SYNOPSIS AND CONCLUSIONS

In this simulation study, the focus was on analyzing the effects of different distributions of hydraulic parameters on simulated water and solute fluxes. The results demonstrate that the spatial distributions of hydraulic parameters affect the simulated water and solute fluxes. The general patterns of the measured spatial distributions of water fluxes in 110 cm soil depth could be approximated when using spoil cross sections with spatial variability in the hydraulic parameters. However, measured water fluxes exhibited a much stronger heterogeneity than the simulated fluxes, even for the simulation scenarios with small-scale variability in hydraulic parameters (Scenario D). For bromide transport, all simulated scenarios showed less vertical transport and less vertical spreading of the solute plume in comparison with the measured data.

The observed preferential flow process at this site could not be adequately described with any of the simulation scenarios using different spatial distributions of hydraulic parameters. Consequently, whether the real heterogeneity was still underestimated in the simulated scenarios, or if mechanisms other than spatial heterogeneity in hydraulic parameters contributed to the observed preferential flow remains unresolved.

None of the factors, which were incorporated in the simulated scenarios in addition to spatial variability in hydraulic parameters (local scale anisotropy of hydraulic

conductivity, seepage faces between the suction cells, horizontally heterogeneous root distributions) could explain the observed heterogeneity in measured water and solute fluxes satisfactorily, even when all additional factors were combined. The preferential flow phenomena observed experimentally at the Bärenbrücker Höhe site were possibly caused by a combination of mechanisms and not attributable to a single process or other factors not accounted for in the present simulations.

Soil water repellency (Gerke et al., 2001) probably induced hysteresis in the hydraulic functions. There are, however, no experimental data for the wetting retention curve at the Bärenbrück site. Moreover, in flow modeling a different hydraulic conductivity function should be used for the wetting branch (Nieber et al., 2000). This is beyond the scope of the present version of the HYDRUS-2D model. Hysteresis in hydraulic functions should be tested in future studies provided appropriate hydraulic data and model approaches especially designed for flow in hysteretic and hydrophobic soils are available.

Another factor, which could possibly initiate preferential flow at the Bärenbrücker Höhe site is the microtopography of the soil surface (Hangen, 2003). This was also not accounted for in the present simulation study.

The experimental database of hydraulic parameters must be extended with regard to the effect of soil hydrophobicity on the hydraulic functions, and seasonal variability in processes within zones of high root density. Also, use of a single continuum model with spatially distributed hydraulic parameters, as applied here, assumes the existence of local equilibrium. If this assumption is violated, use of dual-permeability models should be considered, in either 1D (e.g., Gerke and van Genuchten, 1993) or 2D (e.g., Vogel et al., 2000).

ACKNOWLEDGMENTS

This project was financially supported by the German Research Foundation (Deutsche Forschungsgemeinschaft—DFG), Bonn through grant GE 990/2-2. We would like to thank several persons who contributed to this work: Dr. M. Einecke and Dr. B. Wecker (from the Chair of Soil Protection and Recultivation, Brandenburg University of Technology, Cottbus) provided data of physical mine soil properties and chemical analyses of mine soil; Dr. E. Hangen (Bavarian Geological Survey, Marktredwitz) measured drainage rates, precipitation data, and the root distribution. The authors would like to thank Prof. Dr. R. F. Hüttl (Chair of Soil Protection and Recultivation, Brandenburg University of Technology at Cottbus) for his stimulating support. We thank two anonymous reviewers, whose comments improved the manuscript. We are very thankful to Dr. M.Th. van Genuchten for his editorial suggestions for improving the readability of the manuscript.

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